NAME: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 4 CHRONOLOGICAL ENQUIRY SEQUENCE • C1900–PRESENT

Medicine in Modern Britain, c1900–present

c1900–present: Ideas on Causes, Diagnostic Technology, The NHS, Penicillin & Lung Cancer



HISTORICAL PRIMARY RECORD • 5 JULY 1948 COI-INF-12-88

Aneurin Bevan with First NHS Patient Sylvia Beckingham, Park Hospital

Central Office of Information • Minister of Health Aneurin Bevan inaugurating the National Health Service at Park Hospital, Davyhulme, marking universal free access to healthcare.

PRIMARY VISUAL EVIDENCE THE NATIONAL ARCHIVES • INF 12/88

PEARSON EDEXCEL OPTION 11 • KEY TOPIC 4 CONTENT OFFICIAL SPECIFICATION DIRECTORY

1 Ideas About the Cause of Disease and Illness

- Advances in understanding the causes of illness and disease: the influence of **genetic and lifestyle factors** on health.
- Improvements in diagnosis: the impact of the availability of **blood tests, scans and monitors**.

2 Approaches to Prevention and Treatment

- The extent of change in care and treatment. The impact of the **NHS** and science and technology: improved access to care; advances in medicines, including **magic bullets and antibiotics**; high-tech medical and surgical treatment in hospitals.
- New approaches to prevention: **mass vaccinations** and government lifestyle campaigns.

3 Case Studies

- Key individuals: **Fleming, Florey and Chain's development of penicillin**.
- The fight against **lung cancer in the twenty-first century**: the use of science and technology in diagnosis and treatment; government action.

EXAM MATRIX EDEXCEL PAPER 1 SECTION B EXAMINATION TARGETING FRAMEWORK • THEMATIC STUDY [36M + 4M SPAG • 55 MINS] SPECIFICATION 1HI0/11

Q3 • 4 MARKS AO1/AO2 CONCEPT DRILL

Similarity or Difference Across Eras

Target: Explain 1 way [ideas / treatments / care] in this era were similar or different to another period.
Rule: Provide specific factual evidence from *both* named periods to secure 4/4 marks (approx. 5 mins).

Practice: In-Workbook 4-Mark Comparative Tasks

Q4 • 12 MARKS AO1/AO2 CAUSATION

Explain Why [Causal Change] Occurred

Target: Multi-causal analytical essay explaining why medical change or continuity took place.
Rule: Write 3 developed PEEL paragraphs. Must address both stimulus points + 1 distinct point of own knowledge (18 mins).

Practice: Chronological Inquiry Spine & Causation Drills

Q5/6 • 16+4 SPaG AO1/AO2/AO4 SYNOPTIC

'How Far Do You Agree?' Evaluative Essay

Target: Choose Q5 OR Q6. Evaluative thematic essay assessing factor significance or change over time.
Rule: Balanced argument (for/against) with sustained criteria and a decisive historical verdict (30 mins).

Practice: Full 16-Mark Extended Writing Frames

Magic Bullets, Mass Antibiotics & Universal Healthcare

REVISION MIND-MAP CANVAS: At the end of each lesson, use the app's *Consolidate Station* to retrieve your keywords and branch out your mind-map across the open page (Scientific & Wartime Drivers to the left • Clinical & State Impact to the right).

◂ EHRLICH 606 & DOMAGK DYES

1909–32 Magic Bullets: Salvarsan & Prontosil (KT 4.3)

TARGETED CHEMICAL CURE ERA ▸

◂ PETRI DISH MOULD DISCOVERY

1928 Alexander Fleming Finds Penicillin (KT 4.4)

PURIFICATION & FUNDING LIMITS ▸

◂ OXFORD TRIALS & US DEEP VATS

1940–44 Florey, Chain & US Mass Production (KT 4.4)

D-DAY SUPPLY & SUPERBUG RISK ▸

◂ BEVERIDGE REPORT & BMA DEAL

1948 Aneurin Bevan & Birth of the NHS (KT 4.3)

FREE UNIVERSAL HEALTHCARE ▸

SPECIFICATION SPINE CONTINUES TO DNA, SCANNERS & LUNG CANCER (PAGE 3) ➔

The 20th century transformed pharmacology from toxic general poisons into targeted magic bullets.

DNA Genetics, High-Tech Scanners & The Fight Against Lung Cancer

REVISION MIND-MAP CANVAS: At the end of each lesson, use the app's *Consolidate Station* to retrieve your keywords and branch out your mind-map across the open page (Genetics & Scanners to the left • Treatments & Prevention Laws to the right).

◂ X-RAY PHOTO 51 & BASE PAIRS

1953 Franklin, Watson & Crick: DNA Helix (KT 4.1)

HEREDITARY DISEASE EXPLAINED ▸

◂ HOUNSFIELD CT & FIBER-OPTICS

1971–90 CT, MRI Scanners & Endoscopy (KT 4.2)

NON-INVASIVE SOFT-TISSUE SCANS ▸

◂ 3 BILLION BASE PAIRS MAPPED

2003 The Human Genome Project Mapped (KT 4.1)

GENE SCREENING VS THERAPY LAG ▸

◂ PET-CT, VATS & IMMUNOTHERAPY

2007–24 Lung Cancer: Oncology & State Bans (KT 4.5 • 4.2)

2007 SMOKING BAN & PLAIN PACKS ▸

KEY TOPIC 4 SPECIFICATION SPINE COMPLETE ➔

Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?

AUDIO • QUIZ

1 c1900–1940s • Why did early doctors struggle to explain hereditary traits?

- Gregor Mendel genetics
- Chromosomes observed
- Unknown hereditary mechanism

2 1951–1952 • How did Photograph 51 prove DNA's helical structure?

- Rosalind Franklin
- X-ray crystallography
- Photograph 51 (1952)

3 1953 • How did base pairing explain hereditary replication?

- Watson & Crick model
- Double helix structure
- DNA base pairing

4 1990–2003 • Why was mapping 3 billion base pairs a medical milestone?

- Human Genome Project
- Mapped 3bn base pairs
- International teamwork

5 2000s–Present • Why did understanding causes outpace finding cures?

- Genetic screening
- BRCA1/2 gene testing
- Gene therapy lag

Franklin's 62-hour X-ray exposure provided the decisive physical evidence that DNA was a double helix.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why there was rapid progress in understanding the causes of disease in the period c1950 to the present. [12 marks]

Stimulus: **Rosalind Franklin's Photograph 51 (1952)** | **The Human Genome Project (1990–2003)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

KEY VOCABULARY BANK: Gregor Mendel • Rosalind Franklin • Photograph 51 (1952) • Watson & Crick • double helix model • Human Genome Project (1990–2003) • gene screening • hereditary disease

Write your response in the space provided below:

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT4.1 on Pages 2–3: map Franklin's Photograph 51 and the double helix (left) against the Human Genome Project and hereditary gene screening (right). **[ ] Done**

Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?

AUDIO • QUIZ

1 1895–1918 • How did X-rays transform surgical trauma diagnosis in WWI?

- Wilhelm Röntgen (1895)
- Marie Curie mobile vans
- Shrapnel detection

2 1930s–1960s • Why was fiber-optic endoscopy superior to exploratory surgery?

- Blood tests & typing
- Centrifuges & biopsy
- Fiber-optic endoscopes

3 1971–1980s • Why did CT scans transform internal soft-tissue diagnosis?

- Godfrey Hounsfield (1971)
- EMI CT body scanning
- MRI magnetic imaging

4 c1950–1980s • What drove the shift from infectious to lifestyle diseases?

- Antibiotics curb fevers
- Higher life expectancy
- Cancers & heart disease

5 1980s–Present • Why do governments favour lifestyle prevention over treatment?

- 1983 Seatbelt legislation
- 2007 Public smoking ban
- Sugar tax & Change4Life

Bertha Röntgen cried "I have seen my death!" upon seeing her bones; doctors saw the dawn of non-invasive diagnosis.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Technological innovations in medical diagnosis were more important than public health lifestyle campaigns in improving health in Britain c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **Wilhelm Röntgen's X-rays (1895)** | **The 2007 smoking ban in public places** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, high-tech diagnosis was vital because... • For example, ... • This was significant because... • On the other hand, lifestyle campaigns mattered because... • In particular, ... • Overall, the most decisive factor was... because...*

KEY VOCABULARY BANK: Wilhelm Röntgen (1895) • X-rays • CT scanner (1971) • Godfrey Hounsfield • MRI • non-invasive imaging • 2007 smoking ban • Change4Life

Write your response in the space provided below:

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT4.2 on Pages 2–3: map Röntgen's X-rays, Hounsfield's CT, and MRI scans (left) against lifestyle campaigns and the 2007 smoking ban (right). **[ ] Done**

Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?

AUDIO • QUIZ

1 1906–1909 • Why was Salvarsan 606 termed the first "magic bullet"?

- Paul Ehrlich research
- Magic bullet concept
- Salvarsan 606 (1909)

2 1932–1935 • How did Prontosil prove synthetic drugs could kill microbes?

- Gerhard Domagk (1932)
- Prontosil red dye
- Sulfa drug revolution

3 1942 • How did WWII unity create momentum for state healthcare?

- 1942 Beveridge Report
- Five Giant Evils
- Post-war welfare state

4 1945–1948 • How did Bevan compromise to persuade resistant doctors?

- Aneurin Bevan (1948)
- BMA doctor compromise
- Free universal healthcare

5 1950s–Present • Why has modern medical success created financial strain?

- Kidney dialysis units
- Hip replacement surgery
- Rising NHS cost pressures

Ehrlich tested 605 failed arsenic compounds before compound 606 finally cured syphilis in 1909.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'The establishment of the National Health Service in 1948 was the most significant turning point in medical care in the period c1900–present.' How far do you agree? [16+4 marks]

Stimulus: **The National Health Service (1948)** | **The discovery of Salvarsan 606 (1909)** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, the NHS was transformative because... • For example, ... • This was significant because... • On the other hand, magic bullets were crucial because... • In particular, ... • Overall, the most decisive factor was... because...*

KEY VOCABULARY BANK: Paul Ehrlich • Salvarsan 606 (1909) • Gerhard Domagk • Prontosil (1932) • William Beveridge (1942) • Aneurin Bevan • NHS (1948) • free at point of delivery

Write your response in the space provided below:

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT4.3 on Pages 2–3: map Ehrlich's Salvarsan 606 and Domagk's Prontosil (left) against the Beveridge Report and Bevan's 1948 NHS launch (right). **[ ] Done**

Why did penicillin require a global war and American industrial might to become a mass-produced cure?

AUDIO • QUIZ

1 1928–1929 • Why could Fleming not turn mould into a clinical medicine?

- Alexander Fleming (1928)
- Penicillium notatum mould
- Staphylococcus halo

2 1938–1940 • How did Florey and Chain overcome the challenge of purifying penicillin?

- Florey & Chain (Oxford)
- Purifying active penicillin
- Norman Heatley apparatus

3 1940–1941 • Why was Alexander's trial both a success and a tragedy?

- 1940 Lethal mice trial
- Albert Alexander (1941)
- Supply exhaustion death

4 1941–1944 • How did US industrial mass-production transform wartime supply?

- US government funding
- Deep fermentation tanks
- 2.3m D-Day doses

5 1944–Present • Why did Fleming warn that antibiotic overuse causes resistance?

- Allied troop survival
- 1945 Nobel Prize
- MRSA superbug threat

Fleming almost washed his petri dish in Lysol, but noticed a clear halo where mould destroyed golden staphylococci.

Question 4: Explain Why [12 marks • 15 mins]

EXTENDED WRITING ASSESSMENT

Explain why penicillin was successfully developed and mass-produced in the period 1928–1945. [12 marks]

Stimulus: **Alexander Fleming's discovery (1928)** | **The impact of the Second World War** | *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *One main reason was that... • For example, ... • This was important because... • Another key factor was... • In particular, ... • Overall, the most decisive reason was... because...*

KEY VOCABULARY BANK: Alexander Fleming (1928) • penicillin • Howard Florey • Ernst Chain • Norman Heatley • Albert Alexander (1941) • deep-tank fermentation • mass production (1944)

Write your response in the space provided below:

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT4.4 on Pages 2–3: map Fleming's 1928 discovery and Oxford purification (left) against US deep-tank fermentation, D-Day doses, and superbugs (right). **[ ] Done**

Why has the 21st-century fight against lung cancer required state coercion rather than high-tech cures?

AUDIO • QUIZ

1 2000s–Present • Why is early lung cancer so difficult to detect?

- Few pain receptors
- Asymptomatic growth
- Late Stage 3/4 detection

2 2000s–Present • How do CT and PET scans improve cancer detection?

- High-resolution CT scans
- PET-CT tracer imaging
- Bronchoscopy & biopsy

3 2000s–Present • How has technology reduced treatment trauma?

- VATS keyhole lobectomy
- Chest cracking avoided
- CyberKnife pinpoint rays

4 2000s–Present • How does immunotherapy harness the immune system?

- Tumour DNA sequencing
- Targeted pharmacotherapy
- Targeted immunotherapy

5 2006–Present • Why has state legislation proved more effective than cure?

- 2007 Public smoking ban
- 2016 Plain packaging
- Smoke-free generation law

Despite high-tech oncology, 10-year lung cancer survival is only ~10%: statutory prevention remains far more powerful than clinical cure.

Question 5/6: Evaluative Essay [16+4 marks • 20 mins]

EXTENDED WRITING ASSESSMENT

'Government action was the main reason for progress in combating lung cancer in the twenty-first century.' How far do you agree? [16+4 marks]

Stimulus: **The Health Act 2006 (2007 public smoking ban)** | **High-tech diagnostic scans and treatments** *(You must also use information of your own)*

ANALYTICAL CONNECTIVES: *On the one hand, government action was decisive because... • For example, ... • This was significant because... • On the other hand, high-tech treatments mattered because... • In particular, ... • Overall, the most decisive factor was... because...*

KEY VOCABULARY BANK: CT scans • PET-CT • bronchoscopy • biopsy • VATS keyhole surgery • CyberKnife radiotherapy • immunotherapy (pembrolizumab) • 2007 public smoking ban • plain packaging (2016) • Pantone 448 C

Write your response in the space provided below:

MIND-MAP MISSION Open the Consolidate Station in the app and branch out KT4.5 on Pages 2–3: map PET-CT scans, VATS keyhole surgery, and immunotherapy (left) against the 2007 smoking ban and 2016 plain packs (right). **[ ] Done**

KEY TOPIC 4 NOTES & EXAMINATION STRATEGY

Notes: Record key causal links, essay plans, and teacher feedback for this unit.

KEY TOPIC 4 NOTES & EXAMINATION STRATEGY

PUPIL NAME: _____ **TARGET GRADE:** ☐

HW DATES & QR	ENQUIRY FOCUS & PARENTAL HOMEWORK VERIFICATION 1. Scan QR to read aloud / track audio narration • 2. Complete 10-Q online quiz • 3. Parent verifies & signs	DO NOW (10M) Topic Recall	Q3 SIM/DIFF	Q4/5 EXTENDED
SET: _____  DUE: _____	**KT4.1: Ideas on Causes: Genetics, DNA & The Human Genome Project** *"Why did the discovery of DNA revolutionize understanding of disease causes, yet take decades to yield treatments?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 4.1 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT4.2: Lifestyle Factors & The Technological Revolution in Diagnosis** *"Were high-tech diagnostic scanners more vital to modern health than state lifestyle campaigns?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 4.2 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT4.3: Magic Bullets, High-Tech Treatments & The Birth of the NHS** *"Was the establishment of the NHS in 1948 more transformative than the development of magic bullets?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 4.3 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT4.4: Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin** *"Why did penicillin require a global war and American industrial might to become a mass-produced cure?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 4.4 RECALL** _____ **/ 10**	_____ **/ 4**	**Q4 Explain** _____ **/ 12**
SET: _____  DUE: _____	**KT4.5: Case Study 2: The Fight Against Lung Cancer (21st Century)** *"Why has the 21st-century fight against lung cancer required state coercion rather than high-tech cures?"* ☐ **Read Aloud / Audio Tracked** **10-Q Online Quiz Score:** _____ **/ 10** **Parent / Carer Signature:** _____	**KT 4.5 RECALL** _____ **/ 10**	_____ **/ 4**	**Q5/6 Essay** _____ **/ 20**

Retain this student assessment record for parent consultations and GCSE revision monitoring.